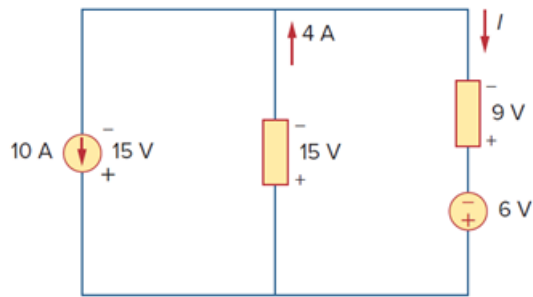


Problem

Find I and the power absorbed by each element in the network of Fig. 1.30.

Figure 1.30:



Step-by-step solution

Step 1 of 5

Refer to circuit in Figure 1.30 in the textbook.

From Figure 1.30 in the textbook, apply Kirchhoff's current law at top node.

$$10 - 4 + I = 0$$

$$I = -6 \text{ A}$$

Therefore, the current I is $\boxed{-6 \text{ A}}$.

Step 2 of 5

Calculate the power absorbed by the elements in the circuit.

$$P = VI$$

Find the power absorbed by the 10 A current source.

Since the current enters at the negative terminal of the 15 V, the current delivers the power to the other elements in the circuit. Hence, it **does not absorb** the power.

The power delivered by the source is,

$$\begin{aligned} P_{10\text{A}} &= -(15)(10) \\ &= -150 \text{ W} \end{aligned}$$

Thus, the power delivered by the source is $\boxed{-150 \text{ W}}$.

Step 3 of 5

Find the power absorbed by the element, whose voltage drop is 15 V.

$$\begin{aligned} P_{15\text{V}} &= (15)(4) \\ &= 60 \text{ W} \end{aligned}$$

Thus, the power absorbed by the element is $\boxed{60 \text{ W}}$.

Step 4 of 5

Find the power absorbed by the element, whose voltage drop is 9 V.

$$\begin{aligned} p_{9V} &= (-9)(I) \\ &= (-9)(-6) \\ &= 54 \text{ W} \end{aligned}$$

Thus, the power absorbed by the element is 54 W.

Step 5 of 5

Find the power absorbed by the 6 V voltage source.

Since the current enters at the positive terminal of the voltage source, the element absorbs the power.

$$\begin{aligned} p_{6V} &= -(6)(-6) \\ &= 36 \text{ W} \end{aligned}$$

Therefore, the power absorbed by the 6 V voltage source is 36 W.